

Qwen3.5-9B UK: gender — every mitigation against the baseline on the same rows

Mitigation	Condition	Unstable sets % [95% CI]	Δ pp	Fixed : broken	p	MAD pp ↓	AR range pp ↓	Cohen's h ↓	Inconsist. % ↓	Hire %	Utility % ↑	Mean FS	Attr. mentioned %
baseline (no mitigation)	implicit	14.22 [11.11, 17.56]				2.29	9.78	0.09	3.26	29.74	83.03	0.64	0.24
LEACE (layer 16)	implicit	10.92 [6.91, 15.48] (own baseline 23.14, n=229)	-12.23	44 : 16	4e-04	2.90 (2.29)	13.00 (9.78)	0.14 (0.09)	2.82 (3.26)	41.11 (29.74)	76.51 (83.03)	0.64 (0.64)	0.13 (0.24)
SFT (LoRA, merged)	implicit	3.11 [1.56, 4.89] ♦	-11.11 ♦	62 : 12	3e-09	0.29 (2.29) ♦	1.33 (9.78) ♦	0.01 (0.09) ♦	0.66 (3.26) ♦	10.99 (29.74)	74.92 (83.03)	0.65 (0.64)	0.09 (0.24)
prompt: ignore_personal_info	implicit	12.44 [9.56, 15.56]	-1.78	20 : 12	0.22	1.29 (2.29)	5.78 (9.78)	0.03 (0.09)	2.53 (3.26)	30.20 (29.74)	82.93 (83.03)	0.64 (0.64)	0.09 (0.24)
scrub: llm	implicit	13.78 [10.00, 17.78]	-0.44	39 : 37	0.91	0.56 (2.29)	2.67 (9.78)	0.03 (0.09)	2.42 (3.26)	29.84 (29.74)	80.91 (83.03)	0.62 (0.64)	0.08 (0.24)
prompt: zero_shot_cot	implicit	14.00 [10.44, 17.56]	-0.22	17 : 16	1.00	1.87 (2.29)	7.78 (9.78)	0.07 (0.09)	2.81 (3.26)	29.28 (29.74)	83.48 (83.03) ♦	0.65 (0.64)	0.21 (0.24)
prompt: reasoning	implicit	14.32 [11.11, 17.90]	+0.00	15 : 15	1.00	1.82 (2.29)	8.89 (9.78)	0.06 (0.09)	2.71 (3.26)	27.32 (29.74)	82.02 (83.03)	0.60 (0.64)	0.26 (0.24)
prompt: structured_rubric	implicit	14.22 [10.67, 17.78]	+0.00	40 : 40	1.00	2.11 (2.29)	10.89 (9.78)	0.07 (0.09)	2.83 (3.26)	19.03 (29.74)	77.79 (83.03)	0.56 (0.64)	0.07 (0.24)
prompt: recruiter_guidelines	implicit	14.25 [10.89, 17.86]	+0.22	21 : 22	1.00	1.94 (2.29)	8.67 (9.78)	0.05 (0.09)	3.07 (3.26)	26.84 (29.74)	82.18 (83.03)	0.64 (0.64)	0.19 (0.24)
prompt: fairness_constitution	implicit	14.70 [11.16, 18.49]	+0.45	33 : 35	0.90	1.89 (2.29)	6.44 (9.78)	0.07 (0.09)	3.00 (3.26)	35.66 (29.74)	80.66 (83.03)	0.66 (0.64)	0.06 (0.24) ♦
prompt: counterfactual_invariance	implicit	18.75 [14.80, 22.94]	+4.69	21 : 42	0.01	2.46 (2.29)	10.44 (9.78)	0.08 (0.09)	3.35 (3.26)	33.30 (29.74)	81.26 (83.03)	0.63 (0.64)	0.13 (0.24)
prompt: second_pass_verification	implicit	29.31 [24.50, 34.16]	+15.44	13 : 82	2e-13	3.89 (2.29)	15.78 (9.78)	0.13 (0.09)	6.91 (3.26)	27.94 (29.74)	78.63 (83.03)	0.63 (0.64)	0.20 (0.24)
scrub: lexical*	implicit	0.44 [0.00, 1.11]	-13.78	63 : 1	7e-18	0.09 (2.29)	0.44 (9.78)	0.00 (0.09)	0.09 (3.26)	28.89 (29.74)	81.78 (83.03)	0.63 (0.64)	0.10 (0.24)
baseline (no mitigation)	explicit	33.78 [28.00, 39.56]				3.73	30.44	0.09	4.36	30.31	80.11	0.62	1.80
scrub: llm	explicit	0.89 [0.22, 1.78] ♦	-32.89 ♦	150 : 2	4e-42	0.14 (3.73) ♦	0.44 (30.44) ♦	0.00 (0.09) ♦	0.20 (4.36) ♦	31.04 (30.31)	80.73 (80.11) ♦	0.62 (0.62)	0.06 (1.80) ♦
SFT (LoRA, merged)	explicit	4.00 [2.22, 6.00]	-29.78	139 : 5	4e-35	0.52 (3.73)	2.67 (30.44)	0.02 (0.09)	0.81 (4.36)	11.57 (30.31)	75.17 (80.11)	0.65 (0.62)	0.09 (1.80)
LEACE (layer 16)	explicit	31.31 [24.88, 38.39] (own baseline 47.66, n=214)	-16.36	54 : 19	5e-05	4.50 (3.73)	37.31 (30.44)	0.11 (0.09)	4.20 (4.36)	43.67 (30.31)	73.52 (80.11)	0.62 (0.62)	0.54 (1.80)
prompt: fairness_constitution	explicit	20.49 [16.26, 24.89]	-13.36	99 : 39	3e-07	2.55 (3.73)	13.93 (30.44)	0.05 (0.09)	4.08 (4.36)	38.90 (30.31)	80.14 (80.11)	0.65 (0.62)	0.12 (1.80)
prompt: structured_rubric	explicit	23.56 [18.67, 28.44]	-10.22	67 : 21	9e-07	3.17 (3.73)	22.44 (30.44)	0.09 (0.09)	3.48 (4.36)	19.86 (30.31)	77.12 (80.11)	0.56 (0.62)	0.60 (1.80)
prompt: counterfactual_invariance	explicit	25.56 [20.89, 30.67]	-8.22	55 : 18	2e-05	3.36 (3.73)	22.22 (30.44)	0.09 (0.09)	4.26 (4.36)	31.01 (30.31)	79.72 (80.11)	0.62 (0.62)	0.22 (1.80)
prompt: ignore_personal_info	explicit	25.78 [21.11, 30.89]	-8.00	45 : 9	7e-07	2.68 (3.73)	21.78 (30.44)	0.08 (0.09)	3.69 (4.36)	31.42 (30.31)	80.53 (80.11)	0.62 (0.62)	0.84 (1.80)
prompt: recruiter_guidelines	explicit	29.18 [23.78, 34.67]	-4.45	37 : 17	9e-03	3.61 (3.73)	27.56 (30.44)	0.09 (0.09)	4.18 (4.36)	25.39 (30.31)	80.51 (80.11)	0.63 (0.62)	1.67 (1.80)
prompt: reasoning	explicit	31.85 [26.28, 37.36]	-1.78	26 : 18	0.29	3.90 (3.73)	31.11 (30.44)	0.11 (0.09)	4.48 (4.36)	27.31 (30.31)	80.34 (80.11)	0.59 (0.62)	2.41 (1.80)
prompt: zero_shot_cot	explicit	31.85 [26.56, 37.20]	-1.78	27 : 19	0.30	3.58 (3.73)	30.22 (30.44)	0.09 (0.09)	4.12 (4.36)	28.09 (30.31)	80.68 (80.11)	0.64 (0.62)	1.49 (1.80)
prompt: second_pass_verification	explicit	42.83 [37.19, 48.42]	+9.42	37 : 79	1e-04	5.67 (3.73)	29.30 (30.44)	0.15 (0.09)	8.22 (4.36)	30.06 (30.31)	76.67 (80.11)	0.62 (0.62)	1.12 (1.80)
scrub: lexical*	explicit	0.44 [0.00, 1.11]	-33.33	150 : 0	1e-45	0.08 (3.73)	0.22 (30.44)	0.00 (0.09)	0.06 (4.36)	28.94 (30.31)	81.74 (80.11)	0.63 (0.62)	0.09 (1.80)

Values: mitigated run, baseline in brackets, both on exactly the rows the run evaluated. Unstable sets: share with its own 95% CI (bootstrap over candidates); Δ pp: change against the baseline; bold coloured Δ = p < 0.05 (exact sign test on fixed vs broken sets, uncorrected). A run with parse failures is compared on its complete sets only; its own baseline on those sets is shown under its value. Italic ♦ = best value in the column within its condition (real mitigations only; hire rate and FS have no better direction).

Fixed : broken — each counterfactual set (one candidate-job pair, one condition, all attribute variants) paired with the same set at baseline. Fixed: unstable at baseline (the attribute alone changed the decision), stable after the mitigation. Broken: stable at baseline, unstable after — the mitigation introduced a new dependence on the attribute. Δ unstable sets = (broken – fixed) / sets; e.g. 58 : 57 means the method fixes about as many sets as it breaks.

* Lexical scrubbing is an oracle upper bound, not a comparable mitigation: it removes the attribute with the study's own injection templates, so every variant becomes character-for-character the attribute-free CV (100% of rows) and the model sees one identical prompt — zero instability by construction. Real CVs do not use our templates; read this row only as the ceiling. LLM scrubbing, which must find the attribute itself, is the realistic version.